THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For iron reacting with steam, Student A writes the products as $Fe_3O_4 + H_2$; Student B writes $FeO + H_2$. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

2. The pH of a solution with OH^- concentration higher than H^+ is: 1

(a) less than 7 **(b)** more than 7 **(c)** exactly 7 **(d)** 0

3. Lactic acid is present in: 1

(a) vinegar **(b)** orange **(c)** tamarind **(d)** curd

4. A student states: “Iodine is a non-metal and therefore cannot be lustrous.” The correct response is: 1

(a) Correct
(b) Incorrect — iodine is a lustrous non-metal
(c) Iodine is a metal
(d) Cannot be determined

5. A student states: “Copper displaces silver from silver nitrate solution.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only on heating **(d)** Cannot be determined

6. The number of electrons shared in a double bond is: 1

(a) 4 **(b)** 6 **(c)** 2 **(d)** 8

7. The IUPAC name of $CH_3CH_2CH_2OH$ is: 1

(a) propanoic acid **(b)** propanal **(c)** propanone **(d)** propanol

8. Which of these is not required for photosynthesis? 1

(a) chlorophyll **(b)** sunlight **(c)** carbon dioxide **(d)** oxygen

9. Which plant hormone is mostly responsible for closing of stomata in drought? 1

(a) cytokinin **(b)** auxin **(c)** gibberellin **(d)** abscisic acid

10. Which of these is a bisexual flower? 1

(a) cucumber **(b)** papaya **(c)** watermelon **(d)** hibiscus

11. Which organism reproduces by regeneration? 1

(a) yeast **(b)** Rhizopus **(c)** Planaria **(d)** Amoeba

12. The physical expression of a gene is the: 1

(a) phenotype **(b)** chromosome **(c)** allele **(d)** genotype

13. The term for a pair of contrasting forms of a gene is: 1

(a) chromosomes **(b)** alleles **(c)** genomes **(d)** zygotes

14. An object at 2F of a convex lens gives an image: 1

(a) at F **(b)** at infinity **(c)** between F and O **(d)** at 2F, same size

15. The human eye forms the image of an object at its: 1

(a) retina **(b)** pupil **(c)** cornea **(d)** iris

16. The charge on one electron is: 1

(a) 1.6×10^{-19} C **(b)** 9.1×10^{-31} C **(c)** 1.6×10^{19} C **(d)** 1 C

17. An electric heater rated 1500 W used for 2 h consumes: 1

(a) 0.75 units **(b)** 3 units **(c)** 1.5 units **(d)** 30 units

18. Magnetic field lines are close together near the: 1

(a) far away **(b)** nowhere

(c) poles **(d)** middle of the magnet

19. The first trophic level in an aquatic food chain is usually: 1

(a) small fish **(b)** zooplankton **(c)** phytoplankton **(d)** crab

20. Garden and lake are examples of: 1

(a) ecosystems **(b)** food chains **(c)** decomposers **(d)** trophic levels

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Write two observations when lead nitrate solution is added to potassium iodide solution. 2

22. Name two natural acid–base indicators. 2

23. What would happen if platelets were absent? 2

24. Why do we immediately pull our hand away from a hot object? 2

25. A student says Amoeba reproduces by multiple fission. Justify or refute. 2

26. What is a Punnett square? 2

27. Define the centre of curvature of a spherical mirror. 2

28. What is the potential difference between the live and the neutral wire in India? What is the potential of the neutral wire? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. State three effects of oxidation reactions in everyday life with an example each. 3

30. What are the steps in the extraction of a metal from its ore? Name them in order. 3

31. How do nerve cells communicate with muscle cells? 3

32. Explain the events that follow fertilisation in a flower. 3

33. What is the difference between the near point and far point of an eye? What are their values for a normal eye? 3

34. How does the magnetic field of a circular loop become uniform at its centre? 3

35. Explain why 3
(i) antacid tablets relieve acidity
(ii) toothpaste is basic
(iii) lime is added to acidic soil.

OR

What happens when 3
(i) dilute HCl is added to iron filings
(ii) sodium carbonate solution is added to dilute HCl
(iii) NaOH solution is added to HCl with phenolphthalein?

36. Why is an aquarium called an artificial ecosystem? What are its components? 3

OR

What are the harmful effects of excessive use of pesticides? (three) 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object? 5
(ii) Find the magnification.
(iii) Draw (describe) the ray diagram.

OR

Case study: A student focused a distant tree on a screen using a concave mirror and found a sharp image at 15 cm. 5
(i) What is the focal length?
(ii) Nature of the image?
(iii) Where should an object be placed to get a same-size image?
(iv) Find the image distance for an object at 20 cm.
(v) Is that image real?

38. (i) What are addition and substitution reactions? Give one example each with equations. 5
(ii) Which type is shown by saturated hydrocarbons?

OR

(i) What is catenation? 5
(ii) Why does carbon show it more than silicon?
(iii) Draw the structures of a straight chain, branched chain and ring compound of six carbon atoms.

39. (i) A 4 Ω resistor and a 12 Ω resistor are in parallel, and this combination is connected in series with a 1 Ω resistor and a 6 V battery. Find the total resistance and the current from the battery. 5
(ii) Find the current through the 12 Ω resistor.

OR

(i) What is meant by potential difference? 5

(ii) How much energy is given to each coulomb of charge passing through a 6 V battery?

(iii) Calculate the number of electrons in 1 C.

40. (i) What is nutrition? 5

(ii) Describe the steps of holozoic nutrition in Amoeba with a diagram.

(iii) How is it different from that of Paramecium?

OR

(i) Describe the structure of a nephron. 5

(ii) How is urine formed?

(iii) Name the main excretory product in humans and where it is formed.

END OF QUESTION PAPER **Total: 80 Marks**

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